

Certain normal patterns of cerebral activity stand out from the background and resemble abnormalities. They are called “normal variants” as they have no specific clinical significance. Recognizing normal variants avoids mistaking them for epileptogenic abnormalities. Their correct identification prevents a misdiagnosis of seizures or epilepsy [1].

Normal variants can be classified based on one of three features:

1. sharp waveforms, which should be distinguished from epileptiform abnormalities
2. rhythmic waveforms, which should be distinguished from electrographic seizures
3. posterior waveforms, which should be distinguished from posterior abnormalities

Each variant can be identified based on a typical appearance, location, state, and age group as described below.

Sharp Variants

1. wickets
2. small sharp spikes (SSS)
3. positive bursts of 14 and 6 Hz
4. phantom spikes

Rhythmic Variants

1. frontal arousal rhythms (FAR)
2. rhythmic midtemporal theta of drowsiness (RMTD)
3. subclinical rhythmic electrographic (theta) discharges of adults (SREDA)
4. midline theta
5. mu rhythm
6. texting rhythm

Posterior Variants

1. lambda waves
2. posterior slow waves of youth (PSWY)
3. positive occipital sharp transients of sleep (POSTS)

Sharp Variants

These variants have a pointy (sharp) morphology. The first step is to distinguish this pattern from an epileptic discharge.

Both normal variants and epileptic discharges stand out from the background (appear taller than the adjacent waves). Being cerebral activity, both are often negative (deflect upward) and have a physiologic field [2].

Three key features that differentiate a sharp variant from an interictal epileptic discharge are as follows:

1. Normal variants, unlike epileptic discharges, don't disrupt the background, they may simply appear taller than the adjacent waves.
2. Normal variants, unlike epileptic discharges, have symmetric slopes and don't drop below the baseline.
3. Normal variants, unlike epileptic discharges, are not followed by a slow wave.

Figure 13.1 shows the differences between a sharp variant and an epileptic discharge.

Epileptic Discharge



Sharp Variant |



Figure 13.1 Sharp waves (epileptic discharge vs. sharp variant).

The next step is to identify the individual variant. This is done based on its appearance, location, state, and age group along with other characteristics.

Wicket Waves

Appearance: These occur in clusters (trains) with a characteristic arch shape. Single sharps also occur, but their morphology matches those in the cluster. Their amplitudes range from 60–200 μV with typical frequencies of 6–11 Hz that may resemble a Mu rhythm (described below).

Location: anterior to midtemporal, typically bilateral, and independent with shifting predominance.

State: drowsiness and light sleep

Age Group: adults

Clinical Caveat: Wicket waves are commonly misinterpreted as epileptiform abnormalities; this leads to an erroneous diagnosis of epilepsy [3]. Figure 13.2 shows a cluster of wicket waves, Figure 13.3 shows single wicket waves over the left temporal region in the same patient.

Small Sharp Spikes (SSS)

Synonyms: benign epileptiform transients of sleep, benign sporadic sleep spikes (BSSS)

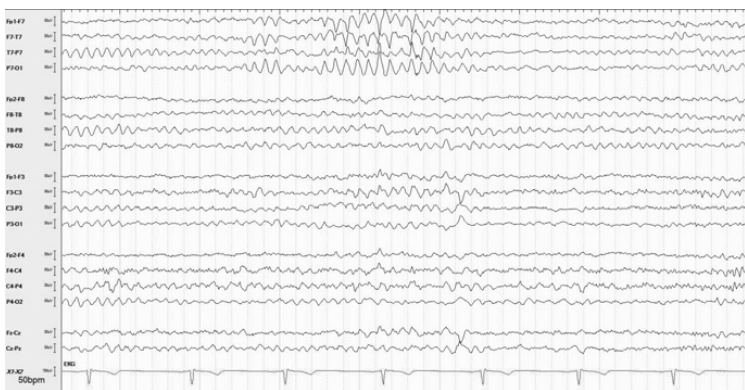


Figure 13.2 Fifty-year-old man with headache; note the cluster of wicket waves over the left temporal region.

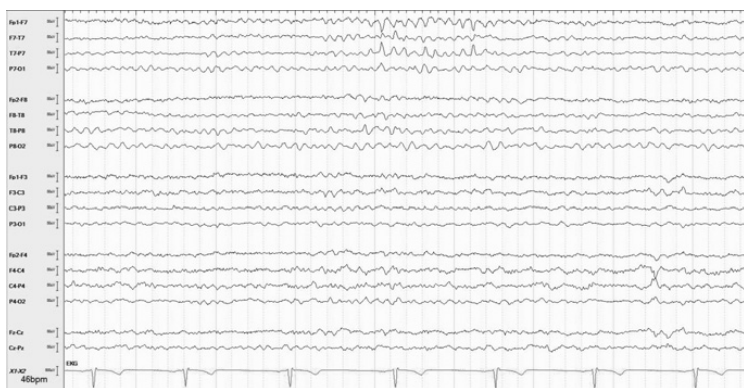


Figure 13.3 Single wicket waves over the left temporal region in the same patient.

Appearance: These occur in singles (never in trains) with a characteristic low amplitude (<50 μ V) and short duration (<50 ms) with near symmetric slopes, they may be mono or diphasic and followed by a small dip in the background. They are not associated with prominent aftercoming slow waves or independent ipsilateral slowing.

Location: temporal (usually bilateral)

State: drowsiness and light sleep (disappear with deep sleep)

Age Group: adults

Best Montage: They have broad fields, best appreciated in montages with long interelectrode distances such as a contralateral ear reference [4]. Figure 13.4 shows small sharp spikes in longitudinal bipolar montage, Figure 13.5 shows the same waveforms in ear reference.

Positive Bursts of 14 Hz and 6 Hz

Synonyms: 14 and 6 Hz positive spikes, ctenoids

Appearance: Brief bursts (0.5–1 s) of alternating positive (downward) spikes and negative (upward) smooth, rounded, comb-like waveforms that resemble a sleep spindle. They may occur at 14 Hz independently, or in association with 6 Hz.

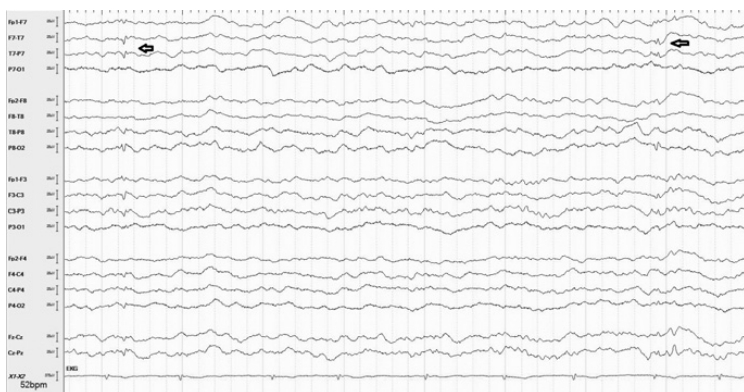


Figure 13.4 Sixty-two-year-old man with mild cognitive symptoms, small sharp spikes (SSS) during light sleep.

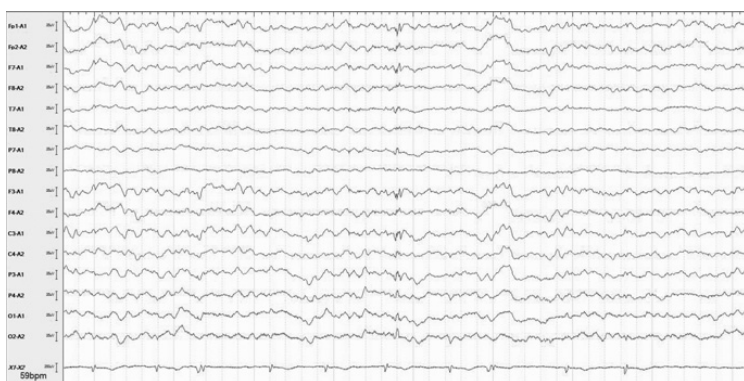


Figure 13.5 Same waveforms as in Figure 13.4 seen in ear reference.

Location: posterior temporal (shifting predominance)

State: drowsiness

Age Group: adolescents (6 to 14 years)

Best Montage: They have broad fields, best appreciated in montages with long interelectrode distances such as a contralateral ear reference [4]. Figure 13.6

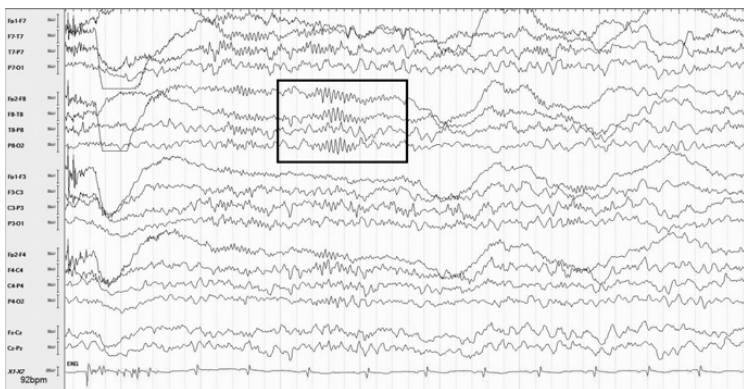


Figure 13.6 Eighteen-year-old man with staring spells, 14 Hz positive bursts with a characteristic spindle-like appearance during wakefulness.

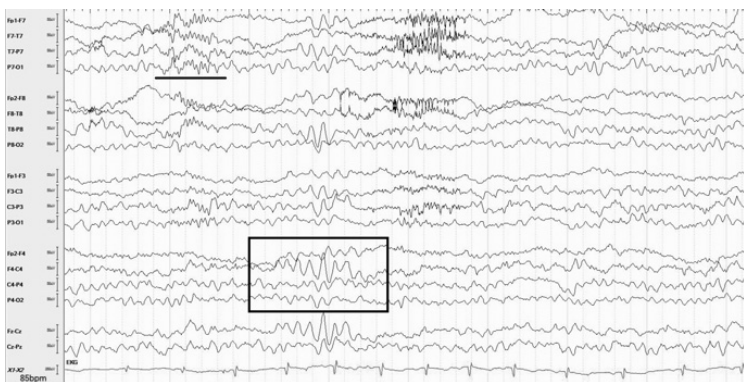


Figure 13.7 14 Hz positive bursts in association with 6 Hz waves in the same patient.

shows 14 Hz positive spikes; Figure 13.7 shows 14 Hz positive spikes in association with 6 Hz waveforms.

Phantom Spike and Wave

Synonyms: 6 Hz spike and wave discharges

Appearance: Brief (1–2 s) synchronous bursts of 5–7 Hz of tiny spikes (called phantoms as they are difficult to see) with higher amplitude slow waves.

Location: generalized, often with an anterior or posterior predominance

State: drowsiness (disappear with deep sleep)

Age Group: adolescents and adults

Best Montage: They have broad fields, so best appreciated in montages with long interelectrode distances such as a contralateral ear reference.

Clinical Caveat: There are two subtypes known by their acronyms. WHAMS (wakefulness, high amplitude, anterior, male) and FOLD (female, occipital, low amplitude, drowsy). The clinical relevance of this differentiation is unclear, however certain features such as higher amplitudes, slower frequencies (<6 Hz), occurrence during wakefulness and persistence in deep sleep suggest a higher association with seizures [4]. Figure 13.8 shows phantom spike and wave in the longitudinal bipolar montage, Figure 13.9 shows the same pattern in ear reference.

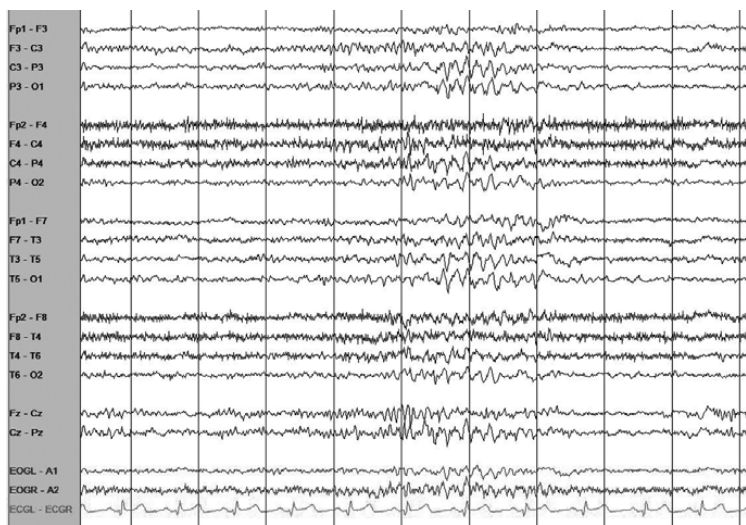
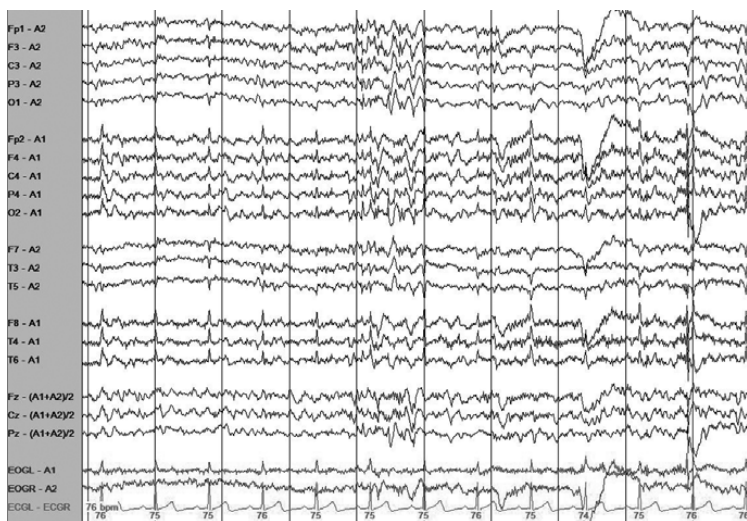


Figure 13.8 Twenty-three-year-old woman with anxiety, phantom spike, and waves most prominent over the occipital channels.



These patterns consist of intermittent bursts of rhythmic activity; hence they can be mistaken for an electrographic seizure.

1. Rhythmic variants have frequencies of 6 Hz or greater which do not evolve (i.e., change with time or in spatial distribution). Epileptic activity including electrographic seizures commonly speed up (crescendo) and less commonly slow down (decrescendo).
2. They disappear in deeper stages of sleep, unlike interictal epileptic seizures which are accentuated during deep sleep.
3. They are always asymptomatic.

Appearance: High-amplitude rhythmic activity of 3–7 Hz occurring with arousal from sleep. May last for as long as 20 s and appear notched due

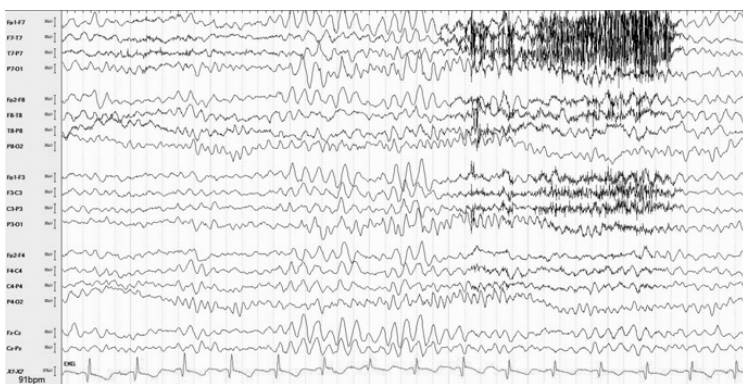


Figure 13.10 Twelve-year-old boy with headache, frontal arousal rhythm (FAR) followed by burst of muscle artifact.

to superimposed harmonics. There is a gradual increase or decrease in amplitude [4].

Location: frontal

State: arousal from sleep, disappears with wakefulness

Age Group: children

Figure 13.10 shows prominent frontal arousal rhythm, followed by muscle artifact indicative of arousal/movement.

Rhythmic Midtemporal Theta of Drowsiness (RMTD)

Synonyms: rhythmic temporal theta bursts of drowsiness, rhythmic midtemporal discharges, psychomotor variant

Appearance: Monomorphic bursts or runs of 5 to 7 Hz theta that appear sharp, flat topped or notched activity that does not show spatial or temporal evolution [4].

Location: temporal (unilateral, bilateral, or shifting)

State: drowsiness

Age Group: adolescents and adults

Figure 13.11 shows a brief left temporal RMTD; Figure 13.12 shows bitemporal RMTD with longer duration over the right temporal region.



Figure 13.11 Forty-year-old woman being evaluated for seizures, burst of left temporal rhythmic midtemporal theta of drowsiness (RMTD) with a notched appearance.

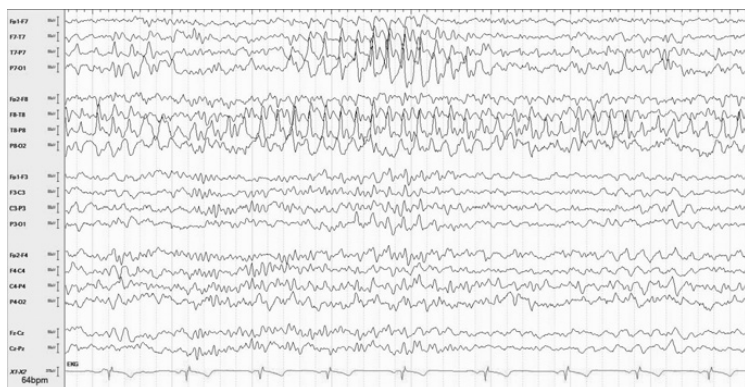


Figure 13.12 Eight-year-old boy being evaluated for seizures, bitemporal RMTD with a sharp appearance, prolonged over the right temporal region.

Subclinical Rhythmic Electrographic (Theta) Discharges of Adults (SREDA)

Appearance: Paroxysmal bursts or runs of rhythmic sharply contoured 5–7 Hz theta that is usually bilateral with a widespread distribution. Focal and unilateral forms also occur. Average duration is a little over a minute. Usually, the onset is abrupt with the page suddenly filled with repetitive monophasic sharp waveforms (Figure 13.13). It may begin with recurring semiperiodic sharp waves (Figure 13.14), at progressively shorter intervals that merge into a rhythmic pattern, usually focal over the central region (Figure 13.15). The offset, as in the onset may be abrupt (Figure 13.16) [4].

Location: Widespread with maximal amplitude over the parietal and posterior temporal region; focal and unilateral forms also occur.

State: drowsiness

Age Group: adults (typically over 50)



Figure 13.13 Sixty-two-year-old woman being evaluated for seizures, onset (abrupt) of diffuse sharp waveforms, characteristic of subclinical rhythmic electrographic (theta) discharges of adults (SREDA).

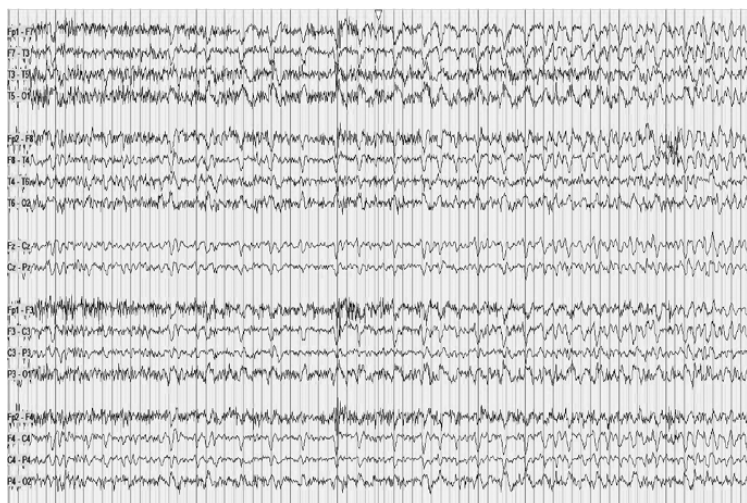


Figure 13.14 Diffuse semiperiodic sharp waves of SREDA in the same patient.



Figure 13.15 Rhythmic SREDA as the pattern progresses in the same patient.



Figure 13.16 Abrupt offset of SREDA in the same patient.

Midline Theta

Synonyms: Ciganek

Appearance: Trains of focal central (vertex) rhythmic theta activity (5–7 Hz) with usually a smooth or sinusoidal appearance but can sometimes be sharply contoured or arciform. Its duration is variable, but typically waxes and wanes [4].

Location: midline (central)

State: wakefulness and drowsiness

Age Group: adults

Figure 13.17 shows a midline theta rhythm.

Mu Rhythm

Appearance: Often asymmetric and asynchronous rhythmic central alpha activity (8–10 Hz) with an arciform morphology.

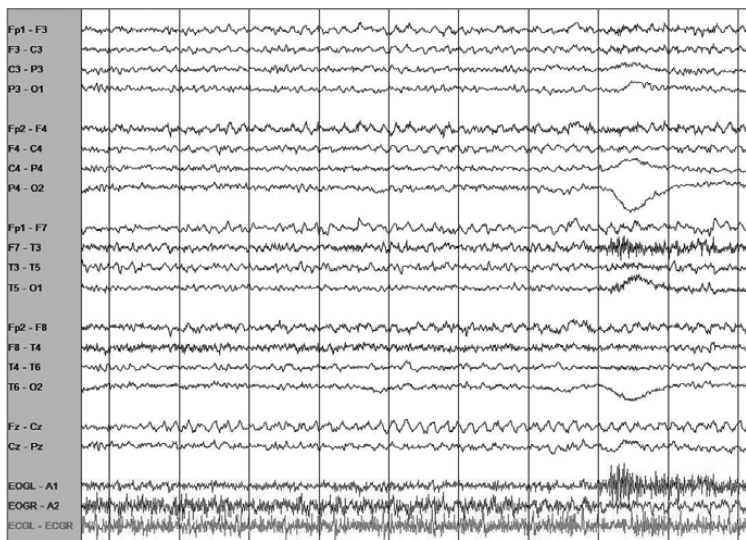


Figure 13.17 Forty-nine-year-old man with episodic movements, central midline theta (Ciganek) rhythm most prominent over the central channels (Cz).

Location: midline (central)

State: wakefulness

Age Group: adolescents and adults

Clinical Caveat: This is the resting rhythm of the sensorimotor cortex, that unlike alpha does not block with eye opening but will block with movement of the opposite extremities [4]. Figure 13.18 shows a Mu rhythm at C3 in longitudinal bipolar montage, Figure 13.19 shows the same waveform in referential montage.

Texting Rhythm

This is a recently described variant characterized by bursts of generalized rhythmic 5–6 Hz monomorphic theta activity predominantly over the frontocentral region and specifically evoked by active phone texting or smart phone use. This may represent activation of cognitive-visual-motor networks during cortical processing [5]. Figure 13.20 shows an example of “texting” rhythm.

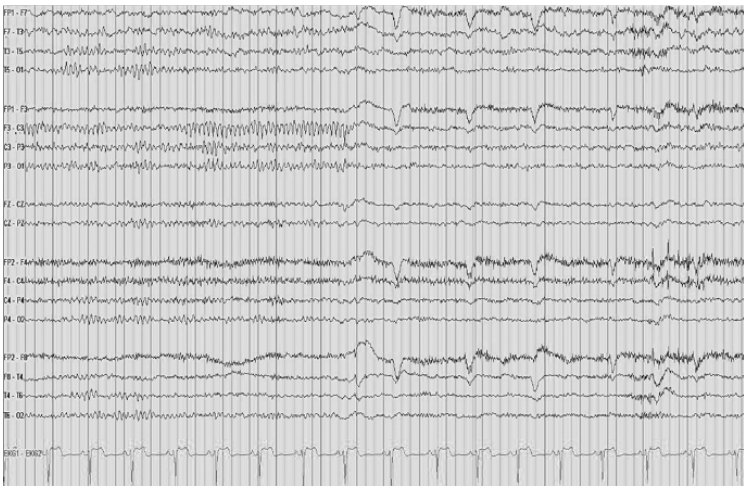


Figure 13.18 Forty-three-year-old man with a craniotomy for resection of a meningioma, prominent Mu rhythm over C3 in the setting of breach effect.

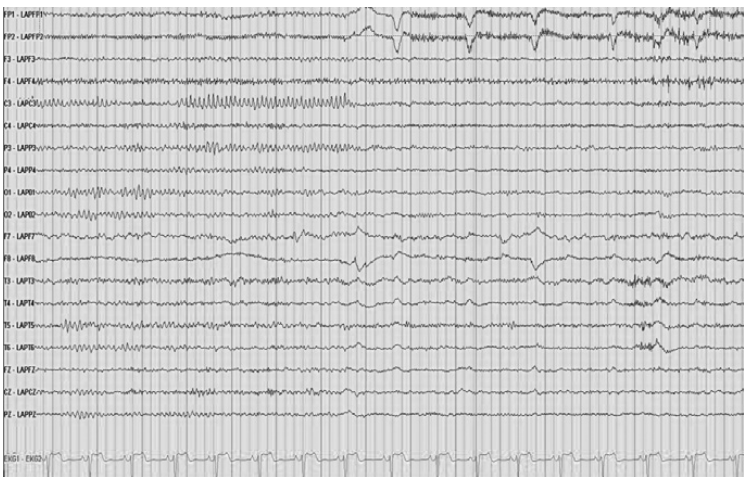


Figure 13.19 Same waveforms seen in Figure 13.18 in average reference montage.

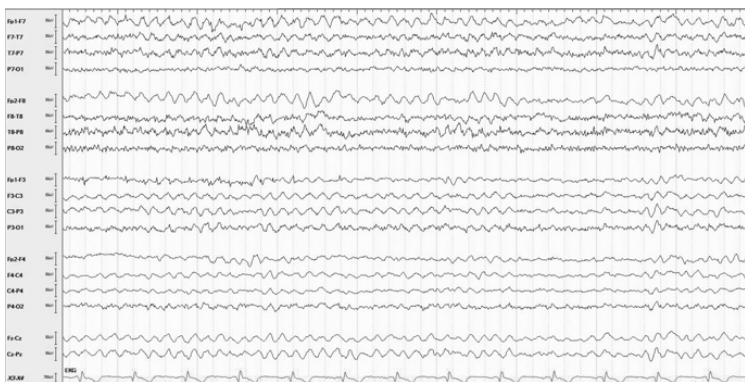


Figure 13.20 Thirty-three-year-old man with shaking spells, frontocentral theta during active smart phone use.

Posterior Variants

These are seen in the occipital channels and occur either as isolated waveforms or trains.

Positive Occipital Sharp Transients of Sleep (POSTS)

Appearance: Positive (downward) occipital sharp waves, often in trains, and commonly asymmetric in amplitude [4].

Location: occipital (phase reverse at O1 or O2)

State: drowsiness and early sleep

Age Group: adolescents and young adults

Clinical Caveat: Confirm positive polarity on a referential montage. Figure 13.21 shows POSTS in drowsiness. Figures 13.22 and 13.23 show POSTS in longitudinal and referential montage, respectively, note the positive deflection in the occipital channels on ear reference (Figure 13.23).

Lambda Waves

Appearance: Look like POSTS, but in wakefulness, and can also be asymmetric.

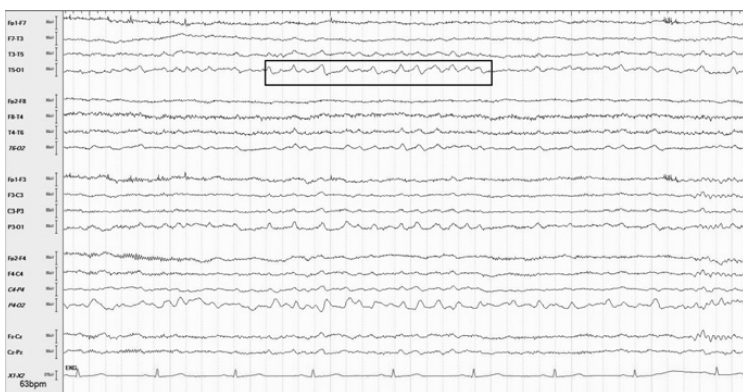


Figure 13.21 Forty-six-year-old man in light sleep, POSTs over the occipital channels.

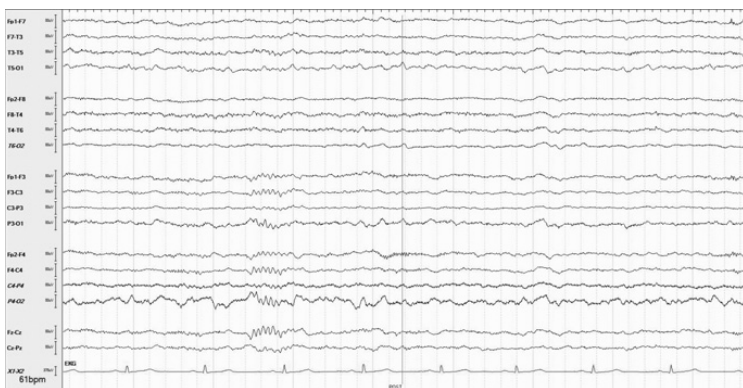


Figure 13.22 Forty-nine-year-old woman in sleep, POSTs in the occipital channels.

Location: occipital

State: wakefulness (active visual exploration)

Age Group: adolescents and young adults

Clinical Caveat: Seen during active visual exploration, hence commonly occur with lateral eye movements. Figure 13.24 shows lambda waves – note features

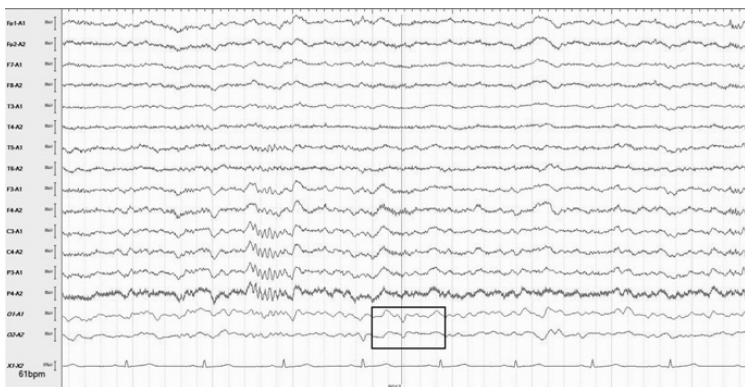


Figure 13.23 Same waveforms as in Figure 13.22 in ear reference; note the downward (positive) deflections definitive for this waveform.

of an “awake” background including eye blink artifact and a reactive posterior-dominant rhythm [6].

Posterior Slow Waves of Youth (PSWY)

Appearance: Often isolated intrusions of broad delta or theta waves (slowing) in the background alpha that are asymmetric and asynchronous [6].

Location: occipital or parietal

State: wakefulness

Age Group: children and young adults

Clinical Caveat: These are reactive to eye opening or alerting stimulation. They increase with hyperventilation. Figure 13.25 shows posterior slow waves of youth.

Notes

1. Some authors believe that occasional focal temporal delta slowing in asymptomatic adults over 60 years of age may be permissible in limited circumstances. *Benign temporal delta transients of the elderly* refers to occasional, small, single, rounded, or irregular delta waves that do not disturb an otherwise normal and symmetric background. These waves are attenuated during wakefulness and accentuated

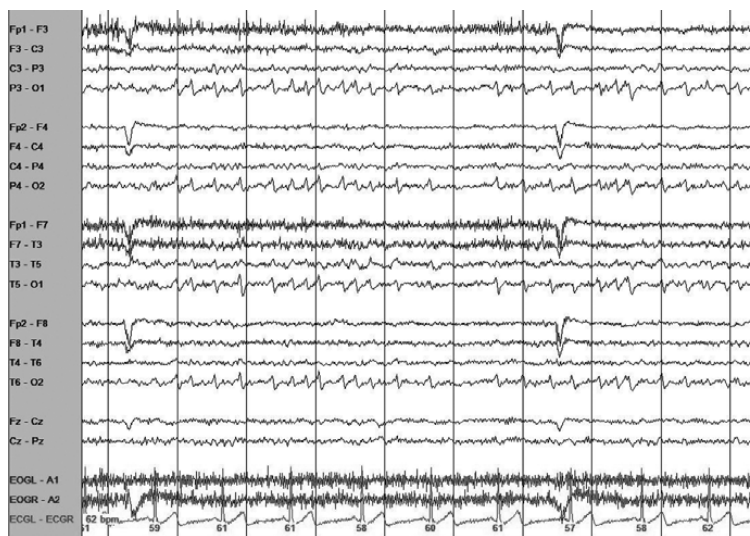


Figure 13.24 Twenty-eight-year-old woman during wakefulness, lambda waves in the occipital channels.

during drowsiness or hyperventilation. Their occurrence should be less than once per ten pages (<1% of the record). They may occur in pairs but never trains [7].

2. Antiepileptic medications such as valproic acid or levetiracetam may blunt epileptic discharges and reduce their amplitude, making them less distinct from the surrounding background, these blunted discharges can be mistaken for variants [8].

Chapter Summary

1. Benign variants should be distinguished from epileptogenic and ictal patterns to prevent misdiagnosis of epilepsy.
2. Variants are identified based on their appearance, typical location, state, and age group.
3. Antiepileptic medications such as valproic acid or levetiracetam may blunt epileptic discharges and reduce their amplitude, making them less distinct from the surrounding background. These can be mistaken for normal variants.

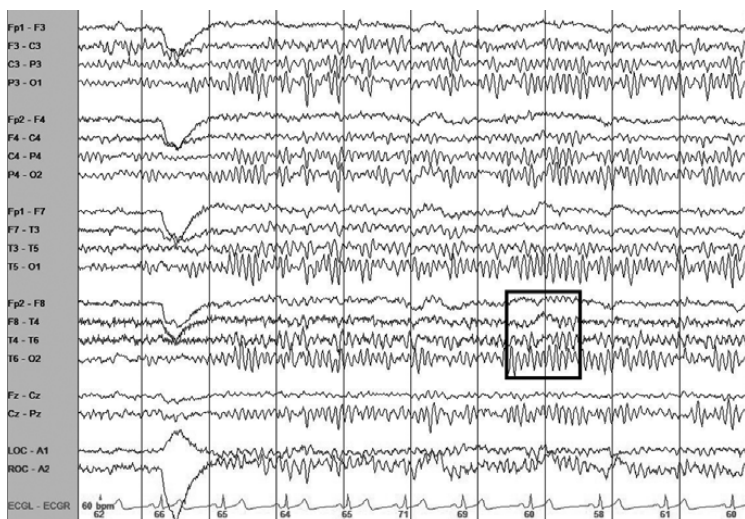


Figure 13.25 Seventeen-year-old girl during wakefulness, posterior slow waves of youth over the posterior temporal region.

Sharp Variants

Variant	Appearance	Location	State	Age group
Wickets	Clusters	Temporal	Drowsiness	Adults
SSS	Singles	Temporal	Drowsiness	Adults
14 and 6 Hz	Bursts	Temporal	Drowsiness	Children
Phantom spikes	Bursts	Diffuse	Drowsiness	Young

Rhythmic Variants

Variant	Frequency	Location	State	Age group
FAR	3–7 Hz	Frontal	Arousal	Children
RMTD	5–7 Hz	Midtemporal	Drowsiness	Adults
SREDA	5–7 Hz	Diffuse	Drowsiness	Adults (>50 years)
Midline theta	5–8 Hz	Midline	Drowsiness	Adults
Mu	8–13 Hz	Central	Awake	Young

Posterior Variants

Variant	Appearance	Location	State	Age group
POSTS	Positive sharps	Occipital	Sleep	Young
LAMBDA	Positive sharps	Occipital	Awake	Young
PSWY	Isolated slowing	Occipital	Awake	Young

References

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